# FlowLedger -- Features & How It Works

Complete Technical and Functional Guide for Users and Developers

## 1. Executive Overview

FlowLedger is a modern, privacy-focused personal expense tracker and financial health monitoring app built with Flutter. It combines friction-free transaction logging via on-device Bank SMS auto-tracking with instant offline SQLite persistence, Cloud Firestore backup, and actionable financial health metrics.

### Core Design Objectives:

- **Zero Cloud Leakage on Ingestion:** Financial SMS detection and regex parsing execute strictly on the client device.
- **Automated Bookkeeping:** Eliminates manual data entry by converting incoming bank notifications into categorized entries.
- **Quantitative Health Scoring:** Provides a standardized 0-100 score based on savings rate and budget discipline.
- **Fluid 60fps Experience:** Optimized widget rebuilds, decoupled frosted glass docks, and snappy swipe physics.

## 2. Bank SMS Auto-Tracking & Parser Engine

FlowLedger integrates an intelligent local SMS listener that automatically logs debits and credits as they happen.

### Operational Flow:

- **Opt-In Security:** Auto-tracking is strictly **OFF by default** on every account. Enabling it triggers an explicit confirmation dialog detailing permissions and on-device processing guarantees.
- **Account Scoping:** Preferences are isolated per authenticated user ID (bank_sms_sync_enabled_[uid]). Switching accounts or logging out immediately terminates the SMS background broadcast listener.
- **Real-Time Toast Feedback:** When a transaction is logged, a floating in-app banner displays the type (Expense/Income), merchant, and parsed amount.

### Parsing Rules & Intelligence (BankSmsParser):

- **Amount Detection:** Supports Indian Rupee formats (Rs., INR, Rs) and decimal values (e.g., Rs. 1,450.00).
- **Debit vs Credit:** Distinguishes expenses ('debited', 'spent', 'paid', 'purchase', 'card') from income ('credited', 'deposited', 'salary', 'refund').
- **Merchant & Category Classification:** Auto-assigns known vendors to relevant categories (Swiggy/Zomato -> Food; Uber/Ola/Petrol -> Transport; Amazon/Flipkart -> Shopping; Electricity/Airtel -> Bills).
- **Spam & Noise Rejection:** Pre-filters OTPs, one-time passwords, login alerts, balance enquiry notices, and marketing messages.

## 3. Financial Health Score (0 to 100 Scale)

Available on the Profile screen, the Financial Health score quantifies a user's current financial discipline and stability by combining monthly cash flow with weekly budget adherence.

| Component | Weight | Mathematical Formula & Thresholds |
|---|---|---|
| **Monthly Cash Flow / Savings Rate** | **70%** | Savings Rate = (Monthly Income - Monthly Expenses) / Monthly Income<br>Score = ((SavingsRate + 0.20) / 0.40 * 100).clamp(0, 100)<br>• **>= +20% Savings Rate:** 100 pts (Full score, standard 50/30/20 personal finance rule)<br>• **0% Savings Rate:** 50 pts (Break-even; spending equals income)<br>• **<= -20% Savings Rate:** 0 pts (Running at a high financial deficit) |

| | | |
|---|---|---|
| **Weekly Budget Discipline** | **30%** | Budget Usage = Weekly Spent on Budgeted Categories / Total Weekly Budget Limit<br>• **Usage <= 80%:** 100 pts (Comfortably within limits)<br>• **Usage >= 120%:** 0 pts (Budget exceeded by 20% or more)<br>• **Usage between 80% and 120%:** Score = ((1.20 - Usage) / 0.40) * 100<br>*Note: If no weekly budgets are configured, the score is 100% based on cash flow.* |

**Score Tiers & Guidance:**

• **75 - 100 ('On a healthy track'):** Excellent savings rate and disciplined weekly spending.

• **50 - 74 ('Room to improve'):** Positive cash flow but narrow savings margin or approaching budget caps.

• **0 - 49 ('Needs attention'):** Outflows exceed inflows or budget thresholds were severely breached.

*If zero income is recorded for the month, the score displays '- / 100' prompting the user to log income.*

## 4. Flow Sense (Heuristic Financial Insights)

Flow Sense provides client-side rule-based financial recommendations directly on the Dashboard:

- **Spending Concentration Analysis:** Identifies which category absorbs the largest percentage of monthly outflow and calculates prospective monthly savings if reduced by 10%.
- **Budget Overrun Detection:** Alerts users immediately when any active weekly budget category crosses 90% or 100% of its allocation, offering actionable adjustment guidance.

## 5. Transaction & Budget Management

The app provides robust ledger management across Activity and Dashboard views:

- **Flexible Grouping & Sorting:** Group transactions chronologically by **Day**, **Month**, or **Year** with ascending/descending order toggles.
- **Type Filtering:** Quick-filter chips for 'All', 'Income', and 'Expense' with dynamic balance totals.
- **Weekly Category Budgets:** Set weekly recurring caps for Food, Transport, Shopping, Bills, etc. Category synonyms are normalized (e.g. Travel is tracked against Transport).
- **Undo-Safe Deletions:** Deleting transactions triggers an instantaneous Undo floating snackbar to prevent accidental record loss.
- **Interactive Calendar:** A monthly view displaying daily spending intensity and monthly totals.

## 6. Architecture, Offline Sync & Customization

FlowLedger is engineered for maximum performance, offline resilience, and aesthetic flexibility:

- **Offline-First SQLite:** Transactions are saved immediately to local SQLite databases, guaranteeing zero latency and full functionality without active internet.
- **Cloud Firestore Sync:** Background synchronization ensures cloud persistence across devices whenever network connectivity is present.
- **Riverpod State Management:** Granular StreamProviders and StateNotifiers prevent unnecessary full-screen rebuilds.
- **Customizable Appearance Engine:**
  - **Dark & Light Modes:** Automatic high-contrast text and surface styling.
  - **Palette Themes:** User-selectable themes (Mint, Ocean, Violet, Sunset).
  - **Ambient Background Toggle:** Soft glowing ambient light or clean solid background.
  - **Frosted Glass Dock:** Adjustable background blur (0-28px) and surface transparency slider.
- **Smooth Swipe Physics:** Custom critically damped spring physics for fluid tab transitions without mid-swipe stutter.

## 7. Security & Privacy Architecture

FlowLedger adheres strictly to local-first privacy standards:

- **On-Device Processing:** Bank SMS content is never transmitted, uploaded, or exposed to third-party servers.
- **Account Isolation:** User images, custom preferences, and SMS auto-tracking flags are isolated per user ID.
- **Secure Auth:** Firebase Authentication powers email/password and Google Sign-In with industry-standard token security.

## 8. Features Summary Matrix

| Feature | Screen / Location | How It Works & Primary Benefit |
|---|---|---|
| **Bank SMS Auto-Tracking** | Settings & Background | On-device SMS listener converts debits and credits into entries with zero manual effort. |
| **Financial Health Score** | Profile Screen | 0-100 metric assessing monthly savings rate (70%) and weekly budget discipline (30%). |
| **Flow Sense** | Dashboard Screen | Heuristic engine highlighting top expense categories and warning of weekly budget breaches. |
| **Weekly Budgets** | Dashboard & Activity | Category-level spending limits with progress indicators and synonym normalization. |
| **Monthly Calendar** | Activity Screen | Date-based ledger view showing daily cash flow distribution and monthly summaries. |
| **Offline SQLite + Cloud Sync** | Core Architecture | Zero-latency offline performance with automatic Cloud Firestore backup when online. |
| **Appearance Customization** | Settings Screen | Dark/light mode, custom color palettes, dock blur slider, and ambient background toggle. |